

NONGBUALAMPHU, Thailand

WMO: 483600

Lat: 17.232N

Lon: 102.429E

Elev: 750

StdP: 14.30

Time Zone: 7.00 (E07)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.1	59.5	44.7	44.9	64.3	48.4	51.7	66.1	13.2	70.6	11.6	71.6	1.8	N/A	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.9	101.3	76.5	98.6	76.4	96.2	76.2	81.0	90.2	80.6	89.4	80.1	88.4	4.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	155.8	83.8	78.7	153.1	83.4	78.2	150.5	83.1	45.6	90.6	44.9	89.6	44.4	88.7	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.2	7.7	51.1	103.6	3.8	3.1	48.3	105.9	46.0	107.7	43.9	109.4	41.1	111.7
			47.8	83.3	3.2	1.5	45.5	84.4	43.6	85.3	41.9	86.1	39.5	87.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	73.3	78.0	82.6	86.2	85.0	84.3	83.0	82.2	82.1	80.9	78.7	73.8
	DBStd	5.40	4.74	4.52	4.87	4.50	3.17	2.46	2.77	2.26	2.07	2.43	3.20	4.82
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	11	6	0	2	0	0	0	0	0	0	0	0	3
	CDD50	11256	722	785	1010	1087	1086	1030	1022	997	963	957	860	737
	CDD65	5793	263	365	547	637	621	580	557	532	513	492	410	276
	CDH74	61602	2107	3773	6487	8279	7264	6805	6036	5330	5088	4632	3605	2196
Wind	WSAvg	2.3	2.9	2.8	2.8	2.5	2.2	1.9	1.9	1.9	1.8	2.0	2.4	2.9
	PrecAvg	60.4	0.4	0.7	1.8	3.3	8.1	9.1	10.0	11.6	10.5	3.8	0.7	0.2
Precipitation	PrecMax	75.9	1.9	3.0	5.6	8.7	12.6	16.9	16.3	22.6	16.9	8.2	3.3	1.7
	PrecMin	44.8	0.0	0.0	0.0	1.1	3.0	3.4	5.3	6.3	4.9	0.1	0.0	0.0
	PrecStd	8.3	0.6	0.7	1.4	1.8	2.6	2.6	2.8	3.6	3.3	2.2	1.0	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.8	97.8	103.2	106.0	101.5	97.8	98.0	93.2	93.5	92.6	93.4	91.5
		MCWB	71.4	72.4	73.6	78.3	79.3	78.6	78.1	79.7	79.5	77.4	76.3	73.6
	2%	DB	88.4	95.1	99.8	102.8	98.1	94.7	93.6	91.4	91.3	90.7	90.9	89.3
		MCWB	71.2	72.3	73.2	77.4	79.0	79.2	78.8	79.3	79.4	76.9	75.8	73.1
	5%	DB	86.2	92.5	96.8	99.7	95.4	92.6	91.0	89.7	89.8	89.3	88.9	86.4
		MCWB	70.9	72.0	73.5	77.3	78.8	79.1	78.9	79.0	79.0	76.5	74.7	72.0
	10%	DB	83.7	89.8	94.2	96.7	92.8	90.7	88.8	87.9	88.1	87.8	87.0	84.0
		MCWB	69.7	72.1	73.8	76.9	78.9	79.0	78.7	78.6	78.8	76.1	73.8	70.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.8	77.2	78.7	81.3	81.7	81.7	81.1	81.3	81.6	80.3	79.1	76.5
		MCDB	85.1	89.9	91.2	100.2	93.3	90.4	88.8	89.2	87.9	87.4	87.5	86.5
	2%	WB	74.1	75.4	77.3	79.5	80.8	80.9	80.3	80.4	80.7	79.3	77.8	75.0
		MCDB	83.2	88.1	89.8	95.8	92.1	89.5	87.6	87.6	87.9	86.0	86.1	84.0
	5%	WB	72.8	74.4	76.4	78.7	80.0	80.2	79.8	79.8	80.0	78.7	76.7	73.7
		MCDB	81.7	86.8	89.0	93.6	90.6	88.5	86.9	86.6	86.8	85.1	84.8	82.2
	10%	WB	71.4	73.4	75.5	78.0	79.4	79.7	79.2	79.2	79.5	78.0	75.6	72.2
		MCDB	80.3	85.4	88.4	91.7	89.2	87.6	85.9	85.6	85.8	84.0	83.5	81.5
Mean Daily Temperature Range	5% DB	MDBR	20.0	22.5	20.7	18.9	15.2	12.8	11.5	11.5	11.9	13.3	16.7	19.2
		MCDBR	22.8	26.6	26.0	23.1	19.1	15.8	15.2	13.9	14.4	15.8	18.7	21.4
		MCWBR	8.1	7.8	6.2	5.8	4.7	4.4	4.4	4.6	4.7	4.8	5.9	7.5
	5% WB	MCDBR	18.3	21.5	20.7	19.8	16.1	13.6	12.6	12.7	12.9	13.5	16.8	19.0
		MCWBR	7.4	7.5	6.4	6.1	5.3	4.7	4.6	4.9	5.0	4.9	6.0	7.4
Clear-Sky Solar Irradiance	taub		0.472	0.578	0.667	0.624	0.502	0.453	0.440	0.455	0.501	0.542	0.464	0.465
	taud		2.089	1.832	1.665	1.808	2.171	2.334	2.367	2.308	2.164	2.006	2.203	2.130
	Ebn at Noon		256	235	219	230	256	266	270	268	257	241	258	254
	Edh at Noon		49	66	81	70	48	41	39	42	48	55	43	46
All-Sky Solar Radiation	RadAvg		1510	1635	1647	1787	1708	1591	1500	1456	1462	1536	1547	1463
	RadStd		97	80	92	77	103	104	91	91	90	106	56	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+212	+214

Nomenclature: See separate page